Visvesvaraya Technological University, Belagavi

Alva's Institute of Engineering and Technology
Moodbidri, Karnataka

Department of Computer Science and Engineering

A PROJECT REPORT ON

CropSure: Early Detection and Management of Crop Diseases and Pest Infestations

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering in Computer Science and Engineering

Submitted by

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Under the guidance of

Mr. Anjith Raja R

Academic Year (2026-2027)

CERTIFICATE



This is to certify that the project work entitled "CropSure: Early Detection and Management of Crop Diseases and Pest Infestations" is a bonafide work carried out by Sahana R K (4AL23CS132) and Prathiksha (4AL23CS109), in partial fulfilment for the award of the degree of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2026-2027. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

_____
Mr. Anjith Raja R
Project Guide

External Viva

Name of the Examiners	Signature with date
1. _____	_____
2. _____	_____

DECLARATION

We, Sahana R K (4AL23CS132) and Prathiksha (4AL23CS109), students of the final year Bachelor of Engineering in Computer Science and Engineering at Alva's Institute of Engineering and Technology, Moodbidri, hereby declare that the project work entitled "CropSure: Early Detection and Management of Crop Diseases and Pest Infestations" has been carried out by us under the guidance of Mr. Anjith Raja R, Department of Computer Science and Engineering, and is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2026-2027.

We further declare that this report has not been submitted previously for the award of any degree, diploma or similar title to any other university or institution. Where the prototype uses simulated predictions or weather data, this is stated clearly in the report.

Place: Mijar
Date: 30/09/2026

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Sahana Ravi Kangal
Prathiksha

ABSTRACT

Crop diseases and pest infestations are often identified only after visible damage has spread. Extension staff cover large areas, laboratory diagnosis takes time, and the data that influences crop-health risk, such as weather, crop stage, variety, soil condition and local pest history, is rarely brought together at farm level. As a result, treatment is delayed, pesticides may be used excessively or inappropriately, cultivation costs rise and yields fall.

This document describes CropSure, an AI-powered crop-health intelligence platform that shifts crop protection from reactive diagnosis to proactive, data-driven guidance. It provides image-based symptom identification with a confidence score, field-level risk forecasting (Low, Medium or High), geospatial hotspot mapping for agriculture officials, expert validation of uncertain or high-risk cases, multilingual advisories based on Integrated Pest and Disease Management (IPDM), and follow-up monitoring after intervention.

The report covers the problem context, requirements, architecture, modules, risk assessment method, data design, the working prototype, testing plan, security and responsible use, expected impact, limitations and future work. The current prototype uses simulated predictions and weather data for demonstration, and this is stated wherever it applies. CropSure is designed to support farmers and experts, not to replace professional agricultural diagnosis.

Keywords: precision agriculture, crop disease detection, risk forecasting, hotspot mapping, expert-in-the-loop, IPDM, multilingual advisory.

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CHAPTER 1: INTRODUCTION

1.1 Background

Agriculture is a sequence of time-sensitive decisions. When to irrigate, when to spray and what to spray depend on how healthy the crop is and how likely it is to become unhealthy in the coming days. A disease that is recognised a week late can spread across a field, and a wrong diagnosis can lead to the wrong treatment. Both outcomes cost money, damage yield and increase the burden on the environment.

Most farmers rely on personal experience, neighbours, input dealers or occasional visits by extension staff. These sources are valuable, but they are not always available at the moment a symptom first appears. At the same time, useful information already exists in separate places: weather forecasts, sensor readings, records of earlier outbreaks in the area and the knowledge of agricultural experts. It is rarely combined into one clear view that a farmer can act on.

CropSure is built around a simple idea: give every field early, local and understandable crop-health guidance. A farmer can scan a crop, see a probable diagnosis with a confidence score, check the forecast risk for the field and receive advice in a familiar language. When the case is uncertain or serious, an expert reviews it before the advisory is finalised. Officials see the wider picture through dashboards and hotspot maps and can plan extension visits where they are needed most.

1.2 Motivation

The motivation for CropSure is to move crop-health management from reactive diagnosis to proactive, data-driven intelligence. The project brings together three ideas that are usually treated separately: detecting problems from images, predicting the risk of problems from context, and involving experts so that advice remains trustworthy.

- **Problems are noticed late.** Symptoms are often recognised only after visible damage has spread across the field.
- **Expert access is limited.** Extension staff cover large areas, and laboratory diagnosis takes time, especially in remote regions.
- **Data is fragmented.** Weather, crop stage, variety, soil condition and local pest history are rarely combined at farm level.
- **Wrong treatment is costly.** Incorrect diagnosis leads to excessive or inappropriate pesticide use, residue concerns and yield loss.
- **Outcomes are not tracked.** Without follow-up, nobody learns whether a treatment worked.

1.3 Objectives

1. Identify probable crop diseases and pest issues from images and report a confidence score with supporting symptoms.
2. Forecast field-level crop-health risk as Low, Medium or High by combining weather, crop, soil and local history data.
3. Map disease occurrence and risk zones so that officials can plan targeted extension visits.
4. Route uncertain and high-risk cases to experts for validation before advisories are confirmed.
5. Provide safe, IPDM-based advisories in the farmer's language.
6. Support follow-up monitoring and feedback after every intervention.
7. Give farmers, experts and administrators dashboards suited to their roles.

1.4 Scope

The scope of this report covers the complete CropSure concept and its working prototype. The prototype demonstrates the full user journey: signing in as a farmer, expert or administrator, registering farms, starting a crop scan, viewing weather and risk information, reviewing cases and monitoring activity from a

dashboard. Predictions and weather in the prototype are simulated for demonstration, and the interface says so. Sensor integration, live weather feeds and large-scale model training are part of the future scope.

The following are outside the scope of the project:

- Replacing professional diagnosis by agricultural experts or laboratories.
- Recommending specific commercial products.
- Financial services such as insurance or credit.

1.5 Key Contributions

- An end-to-end architecture that connects crop scanning, risk forecasting, expert validation, advisory and follow-up.
- A human-in-the-loop design in which uncertain and high-risk cases are always reviewed by an expert.
- An explainable field-level risk score that shows the reasons behind every alert.
- A geospatial hotspot view that helps officials plan extension visits.
- Multilingual advisories based on Integrated Pest and Disease Management (IPDM).
- A working role-based prototype for farmers, experts and administrators, with simulated data clearly labelled.

1.6 Report Organisation

Chapter 2 states the problem and reviews existing approaches, and Chapter 3 lists the requirements. Chapters 4 and 5 cover the architecture and modules. Chapter 6 explains the risk assessment method and Chapter 7 the data design. Chapter 8 describes the technology stack and walks through the prototype using screenshots. Chapters 9 and 10 cover testing, security and responsible use. Chapters 11 to 13 cover expected impact, limitations, future work and the conclusio

CHAPTER 2: PROBLEM STATEMENT AND LITERATURE REVIEW

2.1 Challenges in Crop Health Management

Farmers need timely, reliable and locally relevant crop-health intelligence. Today this need is not met for the following reasons.

- Crop diseases and pest infestations are often identified only after visible damage spreads.
- Extension staff must cover large geographical areas, so visits are infrequent and reactive.
- Laboratory diagnosis and expert advice may not be immediately available, especially in remote areas.
- Weather, crop stage, variety, soil condition and local pest history influence crop-health risk, but they are rarely combined in one place.
- Incorrect diagnosis can cause delayed treatment, excessive or inappropriate pesticide use, higher cultivation costs, residue concerns and yield loss.

2.2 Problem Statement

The problem addressed by this project is: how can a farmer receive an early, reliable and locally relevant assessment of crop health, with expert support when required, and how can agriculture officials see and respond to emerging problems across an area?

2.3 The Conventional Process

In the conventional process the farmer observes symptoms, searches for information manually, waits for an expert, sends a sample to a laboratory when needed, and only then decides on treatment. By the time a decision is made the crop may already be damaged. This chain is slow at every step and offers no view of how the problem is developing in nearby fields.

Table 2.1: Limitations of conventional practice and the proposed response

Limitation	Effect on the farmer	CropSure response
Delayed disease identification	Treatment starts after damage has spread	Image-based screening with confidence score
Limited access to experts	Advice is slow or unavailable	Expert review queue for uncertain and high-risk cases
No localised risk prediction	Preventive action is not possible	Field-level Low, Medium or High risk forecast
Fragmented agricultural data	Weather, soil and history are not used together	Multi-source risk engine
Reactive management	Losses are managed instead of prevented	Early alerts and IPDM-based advisories
Limited follow-up monitoring	The outcome of treatment is not tracked	Follow-up reminders and feedback records

2.4 Review of Existing Approaches

This section describes, in general terms, the approaches that CropSure builds on. It does not claim results from specific published studies.

Image-based plant disease identification. Computer vision models can learn visual patterns such as spots, lesions, discolouration and wilting from labelled leaf images. Such models are useful as a first screening step. Their accuracy depends on the quality and variety of training images, and on how closely field images match the training conditions. Similar-looking symptoms can have different causes, which is why a confidence score and expert review are important.

Weather-driven disease forecasting. Many crop diseases respond strongly to weather. High humidity, leaf wetness, rainfall and particular temperature ranges create conditions in which fungal and bacterial diseases spread quickly, and warm dry spells favour some insect pests. Forecasting models and rule-based advisories use these relationships to warn farmers before symptoms appear.

Integrated Pest and Disease Management (IPDM). IPDM combines monitoring, cultural practices, resistant varieties, biological control and, where necessary, carefully selected chemical inputs. Its goal is to keep losses acceptable while reducing unnecessary pesticide use. CropSure's advisories are designed around this principle, so that the recommended action fits the diagnosis and the risk level instead of defaulting to a spray.

Surveillance and extension services. Agriculture departments track pests and diseases through field visits and reports. Mapping cases by location helps officials see clusters early and direct limited staff to the areas that need attention.

Table 2.2: Comparison of existing approaches and their gaps

Existing approach	Gap	CropSure response
Stand-alone disease-identification apps	Often ignore weather, crop stage and local history	Combines image analysis with field context
Weather-only advisory services	Do not use images or field observations	Adds scans and expert findings to the forecast
Manual surveillance	Slow, and hard to see across regions	Hotspot mapping of confirmed cases
Expert helplines	Depend on staff availability; no structured case history	Structured case queue and stored history

2.5 Research Gap

The gap is not the absence of individual technologies but the absence of a unified, explainable and locally aware workflow. Image-identification tools, weather services, surveillance records and expert helplines each solve one part of the problem. Few connect a diagnosis to field-level risk, route uncertain cases to an expert, and feed confirmed outcomes back into the system. This project therefore focuses on integration, transparency of predictions, and usability for farmers with limited literacy or connectivity.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Stakeholders and Intended Users

Table 3.1: Stakeholders and their needs

Stakeholder	Primary need	Key features used
Farmer	Scan crops, understand risk, receive clear advice and reminders	Crop scan, risk forecast, advisory, alerts
Agriculture expert	Review AI estimates, confirm diagnosis, record field feedback	Expert case review, laboratory referral
Administrator / official	See regional hotspots, case history and surveillance coverage	Dashboards, hotspot map, case history
Laboratory	Receive referrals for confirmation of difficult cases	Referral records

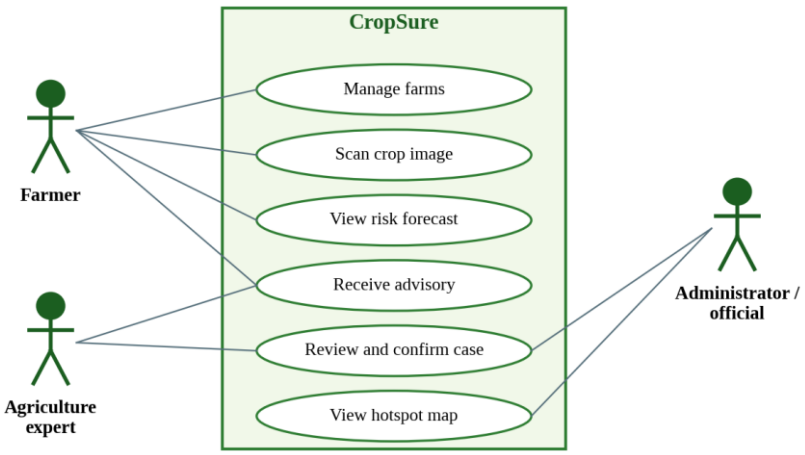


Figure 3.1: Use-case diagram showing actors and system functions

3.2 Functional Requirements

Table 3.2: Functional requirements

ID	Requirement	Module
FR1	Users can register and sign in with role-based access (farmer, expert, administrator).	Access
FR2	Farmers can add and manage farms with location, area, crop variety and crop stage.	Farms
FR3	Farmers can capture or upload a crop image and start a crop scan.	Scan
FR4	The system shows a probable diagnosis, a confidence score and the supporting symptoms.	AI detection
FR5	The system shows a Low, Medium or High risk level for each field.	Risk
FR6	The system combines weather, crop, soil and local history in risk forecasting.	Risk
FR7	Uncertain or high-risk cases can be referred to experts for review.	Expert
FR8	Experts can review AI estimates, confirm diagnoses and record field feedback.	Expert
FR9	The system shows advisories, including safe input recommendations, in the selected language.	Advisory
FR10	Administrators can view case history, hotspots and surveillance coverage.	Admin
FR11	Users receive alerts and follow-up reminders.	Alerts
FR12	Farm data can be linked to sensor readings when sensors are available.	Sensors

3.3 Non-Functional Requirements

Table 3.3: Non-functional requirements

Quality	Requirement
Usability	Simple screens, clear language and large actions suitable for first-time and low-literacy users
Language support	Language selection with regional Indian languages added over time
Performance	Scan results and alerts delivered quickly on typical mobile connections
Reliability	Graceful behaviour when the network or a data source is unavailable
Security and privacy	Role-based access, private storage of farm data and controlled sharing
Transparency	Clear labels wherever data or predictions are simulated
Maintainability	Modular design so that models and data sources can be replaced or improved

CHAPTER 4: SYSTEM ARCHITECTURE

4.1 Design Principles

- **Human in the loop.** Experts confirm uncertain and high-risk cases.
- **Local first.** Risk and advice depend on the farm's own location, crop and stage.
- **Transparent.** Every prediction shows a confidence level and its supporting factors.
- **Modular.** Image models, weather sources and rules can be upgraded independently.
- **Feedback driven.** Expert-confirmed outcomes are stored so that future models can learn from them.

4.2 Layered Architecture

CropSure is designed as a layered system so that each part can be developed, tested and improved on its own. Figure 4.1 shows the five layers and Table 4.1 states the role of each.

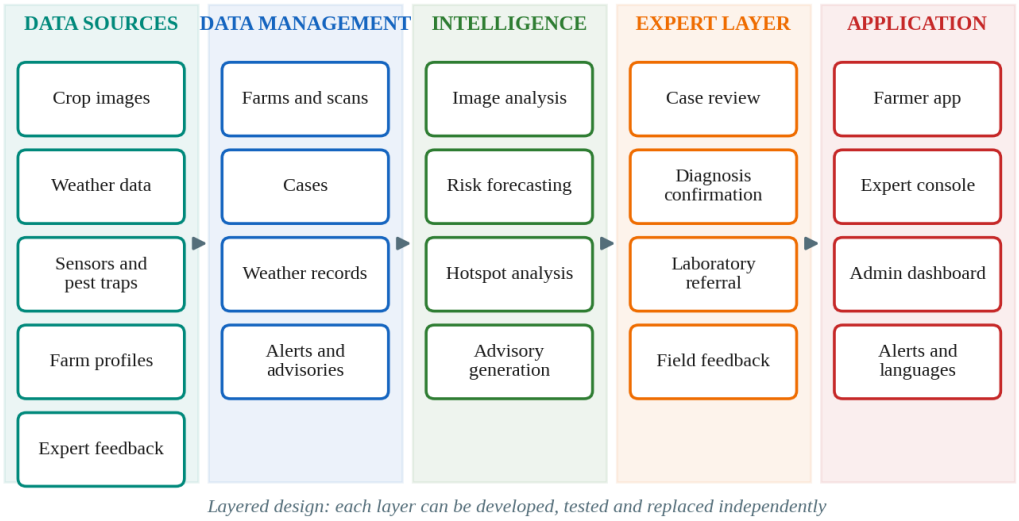


Figure 4.1: Five-layer system architecture

Table 4.1: System layers and their roles

Layer	Role
Data sources	Crop images, weather, sensors and pest traps, farm profiles, expert feedback
Data management	Stores farms, scans, cases, weather records, alerts and advisories
Intelligence	Image analysis, risk forecasting, hotspot analysis and advisory generation
Expert layer	Case review, diagnosis confirmation, laboratory referral and feedback
Application	Farmer, expert and administrator interfaces, alerts and language support

4.3 End-to-End Workflow

A single crop scan passes through ten steps, from image upload to follow-up. Steps 1 to 6 are automatic, step 7 involves an expert only when required, and steps 8 to 10 return guidance to the farmer and record the outcome.

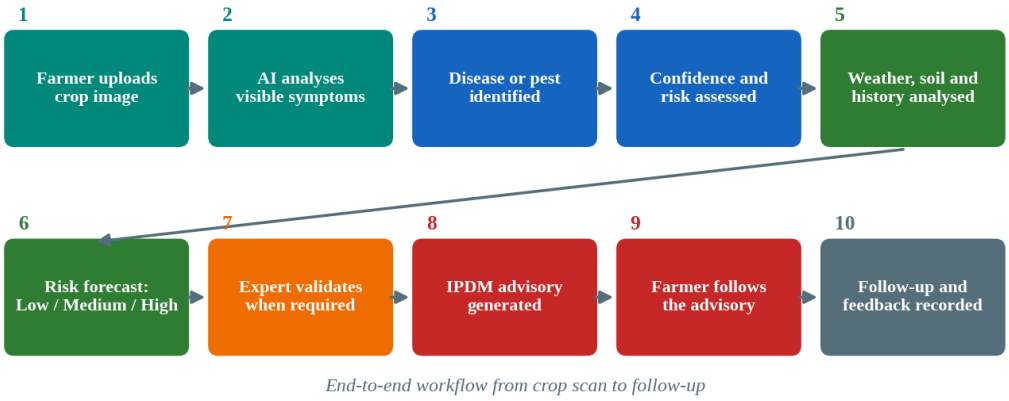


Figure 4.2: End-to-end workflow from crop scan to follow-up

4.4 Data Flow

A crop scan enters the system as an image linked to a farm. The image is analysed and the result is stored as a case. In parallel, the risk engine reads the farm profile, recent and forecast weather, sensor readings and earlier cases in the area. The combined result creates a risk level and an advisory. If the confidence is low or the risk is high, the case appears in the expert review queue. After review, the confirmed diagnosis and advisory are sent to the farmer, and the outcome is stored for follow-up and for improving the models.

CHAPTER 5: MODULES

5.1 AI-Based Crop Disease Detection

The farmer captures or uploads a crop or leaf image. The model analyses the visible symptoms and returns the probable disease or pest issue, a confidence score and the symptoms that support the prediction. If the confidence is low, or the case is high risk, the system recommends expert or laboratory confirmation. Table 5.1 shows an example result.

Table 5.1: Example detection result (illustrative)

Field	Value
Crop	Tomato
Diagnosis	Early Blight
Confidence	85%
Risk	High
Symptoms	Dark concentric leaf spots and yellowing around lesions

5.2 Multi-Source Risk Forecasting

The risk module combines weather, temperature, humidity, rainfall, crop stage, crop variety, soil condition, local pest and disease history, earlier field observations and sensor or pest-trap data. Its output is a risk level and a short explanation, for example elevated humidity and rain in the next two days, so that the farmer understands why an alert was raised.

5.3 Geospatial Hotspot Mapping

Each confirmed case carries a location. The hotspot module aggregates cases by area and time to show disease occurrence, pest hotspots, field-level alerts, regional risk zones and historical patterns. Officials use this view to decide where extension visits will have the most effect.

5.4 Expert Validation and Advisory

The human-in-the-loop flow is shown in Figure 5.1: AI prediction, expert review, diagnosis confirmation, advisory and follow-up. Experts see the image, the AI estimate and the field context. They can confirm or correct the diagnosis, add notes, refer a sample to a laboratory and record field feedback. The advisory that follows recommends safe inputs and IPDM practices in the selected language.



Figure 5.1: Human-in-the-loop validation flow

5.5 Weather and Risk Module

This module shows the local weather outlook and the disease risk forecast for each farm, including the reasons behind the risk score, such as humidity above a threshold or rain expected within 48 hours.

5.6 Sensors Module

The sensors section is designed to bring in readings such as temperature and humidity, and pest-trap data, so that observations from the field can support the forecast. Full sensor integration is part of the future scope.

5.7 Dashboards and Monitoring

Table 5.2: Farmer and administrator dashboard content

Farmer view	Administrator / official view
Crop health status	Farms registered and crop checks
Disease alerts	Farms needing attention
Risk level	Expert reviews awaiting action
Recommended actions	Regional disease hotspots
Weather information	Confirmed diagnoses and case history
Follow-up reminders	Extension response and surveillance coverage

CHAPTER 6: RISK ASSESSMENT METHODOLOGY

The risk engine combines several factors into a single score from 0 to 100, which is then grouped into Low, Medium or High.

> **Note:** The weights and thresholds used in the prototype are illustrative design values. They should be replaced by values calibrated with expert knowledge and real field data before any performance claim is made.

6.1 Input Factors

Table 6.1: Input factors for risk assessment

Factor	Why it matters
Temperature	Each disease and pest has a temperature range in which it spreads fastest
Humidity	High humidity favours many fungal and bacterial diseases
Rainfall	Rain spreads spores and keeps leaves wet
Crop stage	Some stages are more sensitive to specific problems
Crop variety	Varieties differ in tolerance and resistance
Soil condition	Moisture and drainage influence root and soil-borne diseases
Local history	Earlier outbreaks in the area raise the likelihood of recurrence
Field observations	Recent scans and expert findings show what is happening now
Sensor and trap data	Direct readings from the field

6.2 Risk Levels

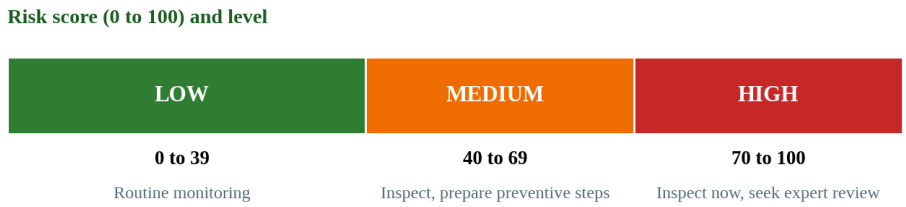


Figure 6.1: Risk score bands and recommended action

Table 6.2: Risk score interpretation

Level	Score range	Typical action
Low	0 to 39	Continue routine monitoring
Medium	40 to 69	Inspect fields, prepare preventive steps
High	70 to 100	Inspect immediately, follow the advisory and seek expert review if symptoms appear

6.3 Explaining the Score

Every risk score is shown with the main reasons behind it. In the prototype, a High risk of 76 out of 100 is explained by humidity above 75% and rain expected within 48 hours. Showing the reasons helps farmers trust and act on the alert, and helps experts judge whether the alert makes sense.

6.4 Calibration and Improvement

The scoring rules should be calibrated in three steps, after which confirmed cases can be used to train a data-driven model that replaces or complements the rules.

1. Agree the initial rules with agricultural experts.
2. Compare alerts with confirmed field outcomes.

3. Adjust the weights and thresholds accordingly.

CHAPTER 7: DATA DESIGN

7.1 Entities

The data design is described at a conceptual level. Table 7.1 lists the main entities and Figure 7.1 shows how they are related.

Table 7.1: Main data entities

Entity	Main information
User	Name, role (farmer, expert, administrator), language, contact details
Farm	Owner, name, location, area, crop, variety, stage, soil details
Crop scan	Farm, image, date and time, model output, confidence score
Case	Scan, diagnosis, risk level, status (AI estimate, under expert review, confirmed)
Expert review	Case, expert, decision, notes, laboratory referral, date
Weather record	Location, date, temperature, humidity, rainfall, forecast
Sensor reading	Farm, sensor type, value, time
Risk assessment	Farm, date, score, level, reasons
Advisory	Case or risk assessment, language, recommended actions, follow-up date
Alert	Recipient, type, message, status

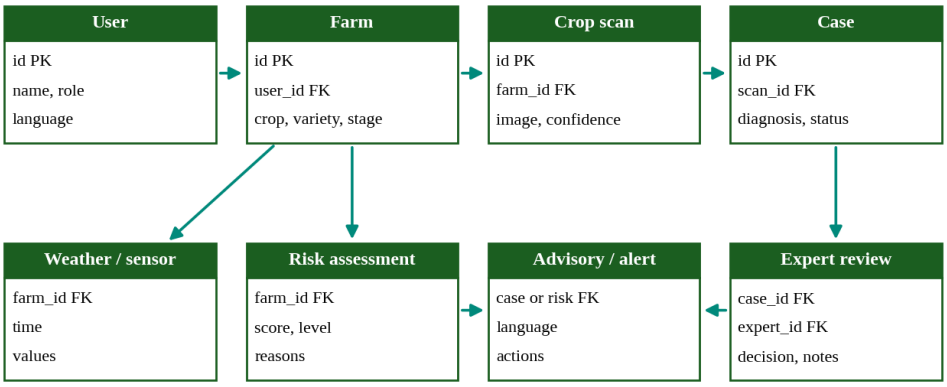


Figure 7.1: Entity-relationship overview of the core data

7.2 Relationships

- A user can own many farms, and a farm can have many scans.
- Each scan produces one case, and a case can have one or more expert reviews.
- A farm has many weather records, sensor readings and risk assessments over time.
- An advisory is linked to a confirmed case or to a risk assessment.

7.3 Data Quality

Images should be stored with the date, location and crop so that they can be used for later training. Expert-confirmed labels should be kept separately from AI estimates, so that model performance can be measured against confirmed outcomes.

CHAPTER 8: IMPLEMENTATION AND PROTOTYPE WALKTHROUGH

8.1 Technology Stack

Table 8.1 lists the technologies used in each layer of the prototype. Weather data is simulated in the current prototype.

Table 8.1: Technology stack of the prototype

Layer	Technology used
Front end	[fill in]
Back end / API	[fill in]
Database	[fill in]
AI / ML model	[fill in]
Weather data source	Simulated in the prototype
Hosting	[fill in]

8.2 Prototype Scope

This chapter shows the working CropSure prototype. The interface uses a calm green palette, plain language and role-based access. A notice on the workspace states that predictions and weather are simulated for demonstration.

8.3 Sign-In Screen

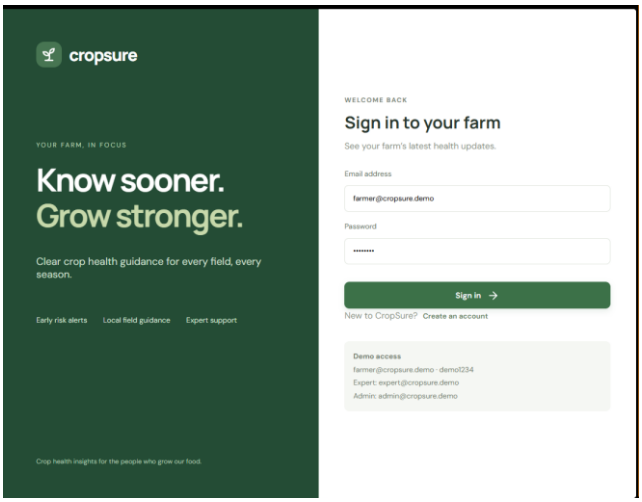


Figure 8.1: CropSure sign-in screen

- The tagline "Know sooner. Grow stronger." states the purpose of the platform.
- Three promises are listed under it: early risk alerts, local field guidance and expert support.
- Users sign in with an email address and password, or create an account.
- Demo access is available for three roles: farmer, expert and administrator.

8.4 Workspace and Expert Case Review

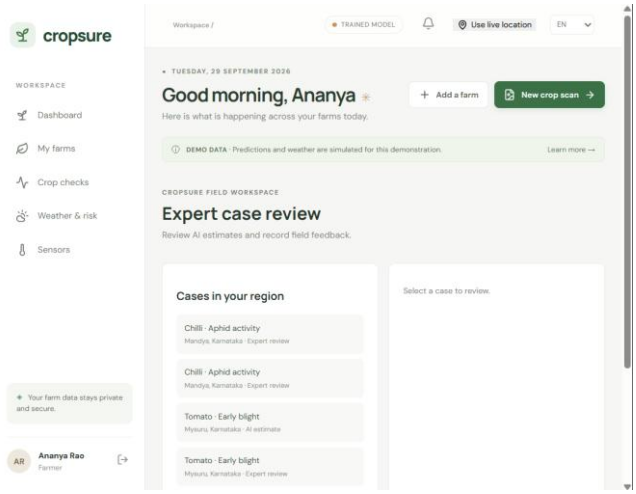


Figure 8.2: Workspace navigation and expert case review

- The left menu contains Dashboard, My farms, Crop checks, Weather & risk and Sensors.
- The top bar shows a Trained model badge, notifications, a live location option and a language selector.
- Add a farm and New crop scan are always available as primary actions.
- The Expert case review panel lists regional cases, for example chilli with aphid activity in Mandya and tomato with early blight in Mysuru, each marked as an AI estimate or awaiting expert review.
- Selecting a case opens it for review so that the expert can confirm the diagnosis and record field feedback.
- A privacy note tells the user that farm data stays private and secure.

8.5 Dashboard

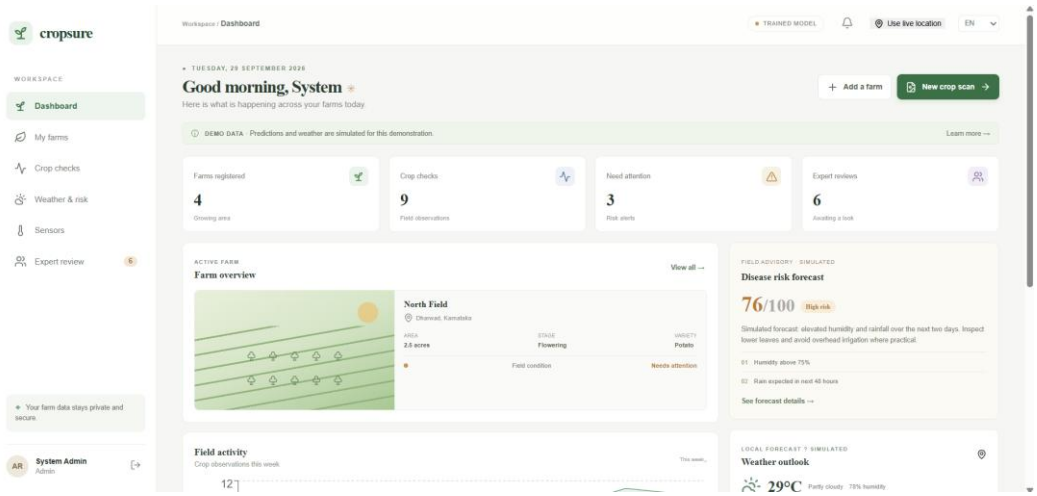


Figure 8.3: Dashboard with farm overview and disease risk forecast

- Summary cards show farms registered, crop checks, farms needing attention and expert reviews awaiting action.
- The farm overview shows the active farm, its location, area, crop stage and variety, and its current condition.
- The disease risk forecast shows a score out of 100 with a risk label and short reasons such as humidity above 75% and rain expected in the next 48 hours.
- The dashboard also shows field activity for the week and a local weather outlook.
- A banner reminds users that the demonstration data is simulated.

The dashboard turns several inputs (farm profile, weather, recent observations and expert queue) into one page, which supports the goal of better farm-level decisions.

CHAPTER 9: TESTING, EVALUATION AND RESULTS

9.1 Testing Plan

Table 9.1: Test types and what is checked

Test type	What is checked
Functional testing	Sign-in, roles, farm creation, scan flow, case review, alerts
Usability testing	Whether farmers can complete a scan and understand the advice unaided
Model evaluation	Accuracy, precision and recall on a held-out set of labelled images
Risk evaluation	Whether alerts agree with confirmed field outcomes
Performance testing	Response time for scans and dashboards
Security testing	Access control and data protection

9.2 Evaluation Metrics

- Detection accuracy, precision and recall for each crop and disease class.
- Share of cases where the confidence score matches expert-confirmed outcomes.
- Time from scan to advisory, and from case referral to expert response.
- Farmer satisfaction and understanding of advisories.

9.3 Results

The prototype currently runs on simulated predictions and weather, so measured performance numbers are not reported here. Table 9.2 is to be completed with measured results once testing is complete.

Table 9.2: Measured results (to be completed after testing)

Metric	Result
Detection accuracy	[00%]
Number of images tested	[000]
Crops and diseases covered	[list]
Average response time	[0 sec]
Farmers or fields piloted	[00]
Expert agreement with AI estimates	[00%]

Observations: [Describe how the system performed, what worked well and what was learned.]

CHAPTER 10: SECURITY, PRIVACY AND RESPONSIBLE USE

10.1 Security and Privacy

- Role-based access limits what farmers, experts and administrators can see and do.
- Farm data stays private to the owner and to authorised reviewers. The prototype states this in its interface.
- Passwords must be stored securely and demonstration credentials must be removed in production.
- Only the minimum data needed for the service should be collected, and users should know how it is used.
- Aggregated hotspot data shown to officials should not expose private details of individual farmers unless they have agreed.

Table 10.1: Risks to users and data, and their mitigations

Risk	Mitigation
Unauthorised access to records	Role-based access for farmers, experts and administrators
Exposure of farm data	Private storage, minimum data collection, controlled sharing
Weak or default credentials	Secure password storage; remove demonstration credentials in production
Identification of individual farmers on maps	Aggregated hotspot data unless the farmer has agreed
Over-reliance on an AI estimate	Confidence score and mandatory expert review for uncertain cases
Simulated data mistaken for real data	Clear labels wherever data or predictions are simulated

10.2 Responsible AI

- CropSure supports experts and farmers and does not replace professional diagnosis.
- Every prediction shows a confidence score, and low-confidence results are routed to experts.
- Simulated data is always labelled as simulated.
- Advisories follow local agricultural department guidelines and label instructions for inputs.
- Recommendations prefer the least harmful effective option, in line with IPDM.

10.3 Fairness and Access

The service should work for smallholders as well as larger farms, on basic smartphones, and in regional languages. Model performance should be checked across crops, regions and image conditions, so that it does not work well for some farmers and poorly for others.

CHAPTER 11: EXPECTED IMPACT AND DEPLOYMENT

11.1 Expected Outcomes

The expected impact chain is: early detection, better decisions, targeted intervention, reduced loss and improved productivity. These are expected benefits that need to be confirmed through field pilots.

- Earlier crop disease detection
- Reduced crop losses
- More targeted pesticide use
- Faster extension response
- Improved surveillance coverage
- Better preventive intervention planning
- Reduced unnecessary cultivation costs
- Improved farm-level decision making
- Better coordination between farmers, experts and laboratories

11.2 Deployment Plan

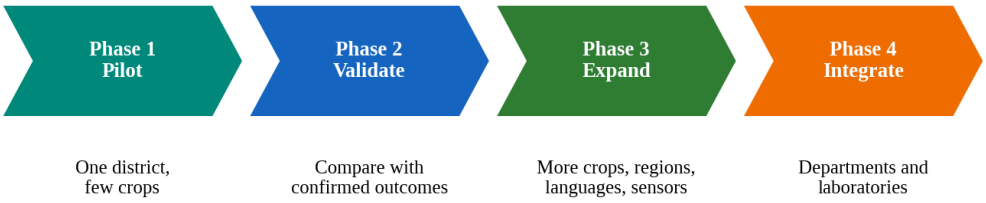


Figure 11.1: Four-phase deployment plan

Table 11.1: Deployment phases and activities

Phase	Activities
Phase 1: Pilot	Select one district and a few crops; onboard farmers and local experts; collect feedback
Phase 2: Validate	Compare alerts and diagnoses with confirmed outcomes; calibrate the risk rules
Phase 3: Expand	Add crops, regions and languages; connect weather and sensor sources
Phase 4: Integrate	Work with agriculture departments and laboratories for surveillance and referral

11.3 Stakeholder Loop

Farmers use the scans and advisories. Experts validate cases. Agriculture departments use hotspot maps and case history to plan extension work. Laboratories receive referrals for confirmation. Together they form a loop in which each confirmed case improves future guidance.

CHAPTER 12: LIMITATIONS, RISKS AND FUTURE WORK

12.1 Limitations

- The current prototype uses simulated predictions and weather, so it has not yet been validated against real field outcomes.
- Model accuracy depends on image quality, lighting and the range of crops and diseases in the training data.
- Visually similar symptoms can have different causes, so expert or laboratory confirmation is needed for uncertain and high-risk cases.
- Risk scores depend on the quality of weather, sensor and history data, which may be incomplete in some areas.
- Regional language coverage is limited at present.
- Network availability can affect uploads and alerts in remote areas.
- Advisories must always be checked against local agricultural guidelines.

12.2 Planned Mitigation

Table 12.1: Limitations and planned mitigation

Limitation	Mitigation
Simulated data	Run field pilots and replace simulated inputs with live data
Limited training data	Collect expert-confirmed images across regions and seasons
Uncertain diagnosis	Confidence thresholds and mandatory expert review
Connectivity	Offline capture with later upload
Language coverage	Add regional languages with expert-reviewed translations

12.3 Future Enhancements

Table 12.2: Future scope

Direction	Description
More crops and disease classes	Extend the model to more crops, pests and nutrient problems
Improved regional models	Train and calibrate models on regional conditions
IoT sensor integration	Bring in soil, temperature and humidity sensors automatically
Automated pest-trap analysis	Count and classify pests from trap images
Satellite and remote sensing	Use vegetation indices to detect stress at scale
More Indian regional languages	Add languages and voice support for easier access
Continuous learning	Learn from expert-confirmed field cases
Department and laboratory integration	Share data and referrals with official systems

Together these steps move CropSure from a demonstration prototype to a dependable field service. The order of work should follow what farmers and experts find most useful during pilots.

CHAPTER 13: CONCLUSION

CropSure transforms crop-health management from reactive diagnosis into proactive, data-driven farm-level intelligence. By combining image-based detection, multi-source risk forecasting, hotspot mapping and expert validation, it gives farmers early and understandable guidance and gives officials a clear picture of what is happening across an area.

The prototype demonstrates the full journey from sign-in to farm overview, crop scanning, risk forecasting and expert case review, with clear labelling of simulated data. The design is also honest about its limits: predictions must be validated on real field data, risk rules must be calibrated with experts, and advisories must follow local agricultural guidelines. The next steps are field validation, calibration with expert knowledge and expansion to more crops, regions and languages.

Summary of Key Points

- Early detection gives farmers time to act before damage spreads.
- Combining images, weather, soil and local history gives a more reliable picture than any single source.
- Expert validation keeps AI advice trustworthy for uncertain and high-risk cases.
- Explained risk scores and plain-language, multilingual advice drive adoption.
- Hotspot maps help officials send limited extension staff where they matter most.
- Field validation and calibration are essential before performance claims are made.

Detect Early. Predict Risk. Act Smart.

REFERENCES

[1] Mohanty, S. P., Hughes, D. P., and Salathe, M. (2016). Using Deep Learning for Image-Based Plant Disease Diagnosis. Frontiers in Plant Science, 7, 1419.

[2] Hughes, D. P., and Salathe, M. (2015). An open access repository of images on plant health to enable the development of mobile disease diagnostics. arXiv:1511.08060.

[3] Kamilaris, A., and Prenafeta-Boldu, F. X. (2018). Deep learning in agriculture: A survey. Computers and Electronics in Agriculture, 147, 70-90.

[4] Food and Agriculture Organization of the United Nations (FAO). Integrated Pest Management (IPM) guidance and resources.

[5] Government of India, Ministry of Agriculture and Farmers Welfare, Directorate of Plant Protection, Quarantine and Storage. Integrated Pest Management guidelines and advisories.

[Add any further datasets, papers, government advisories, tools and libraries used in the prototype.]

Readers should verify the latest editions of guidelines and advisories, which change over time.

APPENDIX A: ILLUSTRATIVE RISK RULE SKETCH

The sketch below shows how factor scores could be combined and grouped into the three risk levels of Table 6.2. The weights are illustrative design values only and must be calibrated with experts and field data. They do not describe measured behaviour.

```
# Illustrative only: weights are placeholders for expert calibration
WEIGHTS = dict(humidity=30, rain_48h=25, temp_range=15,
               crop_stage=15, local_history=15)   # sums to 100

def field_risk(f):
    score = 0
    score += WEIGHTS['humidity']      if f['humidity'] > 75 else 0
    score += WEIGHTS['rain_48h']      if f['rain_expected_48h'] else 0
    score += WEIGHTS['temp_range']    if f['temp_in_favourable_range'] else 0
    score += WEIGHTS['crop_stage']    if f['stage_sensitive'] else 0
    score += WEIGHTS['local_history'] if f['nearby_cases'] > 0 else 0
    return score

def risk_level(score):
    if score >= 70: return 'High'
    if score >= 40: return 'Medium'
    return 'Low'
```

APPENDIX B: SAMPLE CASE RECORD

An illustrative case record, matching the example of Table 5.1, as it could be stored and passed to the expert review queue.

```
{
  "crop": "Tomato",
  "ai_diagnosis": "Early Blight",
  "confidence": 0.85,
  "risk_level": "High",
  "symptoms": ["Dark concentric leaf spots", "Yellowing around lesions"],
  "status": "under_expert_review",
  "language": "selected by farmer",
  "simulated": true
}
```

APPENDIX C: GLOSSARY

Table C.1: Glossary of terms

Term	Meaning
IPDM	Integrated Pest and Disease Management: a balanced approach that combines monitoring, cultural, biological and chemical methods
Confidence score	The model's estimate of how likely its diagnosis is to be correct
Hotspot	An area with a cluster of disease or pest cases
Human in the loop	A design in which experts review and confirm AI output
Extension services	Government or institutional services that advise farmers
Advisory	The recommended actions sent to a farmer for a diagnosis or risk level
Risk score	A value from 0 to 100 that summarises field-level crop-health risk